MODULE 27 · DOMAIN 3 - CLAUDE CODE

Claude Code

The components (CLAUDE.md, rules, Skills, subagents, agent memory), sessions and commands, repository initialisation, auto mode, headless runs in CI, and hooks that help rather than hinder.

DOMAIN 3 of 8 EXAM WEIGHT 3.1% MODULE 27 of 36

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Claude Code is a colleague with a toolbox.
The components decide what it knows, what it can do, and what it must never do.

The components

COMPONENT	WHERE IT LIVES	WHAT IT IS FOR
CLAUDE.md	Managed, user, project, local, subfolders	Context and conventions, concatenated
Rules	.claude/rules/*.md, with paths:	Guidance that loads only for matching files
Skills	.claude/skills, plugins	Procedures; custom slash commands merged in
Subagents	.claude/agents/*.md	Separate contexts for focused tasks
Agent memory	Auto memory; subagent memory field	What Claude learns, kept across sessions
Settings and hooks	.claude/settings.json and scopes	Permissions and enforcement

ON THE EXAM

Only settings and hooks enforce. Everything else is context the model reads.

Skills and commands

- **Custom slash commands are now Skills;** existing command files keep working.
- **Claude invokes a Skill** when its description matches, or you type /skill-name.
- **disable-model-invocation** makes a Skill user-only, for deliberate actions.
- **allowed-tools** narrows what a Skill may use; context: fork runs it in a subagent.
- **Built-in commands:** /init, /clear, /compact, /resume, /memory, /context, /permissions, /hooks, /mcp, /agents.

> THE TIEBREAKER
>
> A Skill the model should never trigger on its own, such as a deploy, sets disable-model-invocation.

Agent memory

- **Auto memory is on by default:** Claude writes what it learns to a project MEMORY.md.
- **Subagent memory:** a memory field at user, project or local scope.
- **Loaded at start:** the first 200 lines or 25 KB of MEMORY.md.
- **Review it with /memory;** edit or delete what is wrong.
- **Memory is not policy.** Shared conventions belong in CLAUDE.md, reviewed.

> THE TAKEAWAY
>
> CLAUDE.md is what the team decided. Memory is what Claude noticed. Promote the useful parts deliberately.

Sessions and starting out

NEED	USE
Start a new repository with Claude	/init: drafts a CLAUDE.md, or suggests improvements
A fresh start on a new task	/clear: empty context
A long session getting crowded	/compact, with optional instructions on what to keep
Pick up yesterday's work	--continue for the latest; --resume for a chosen session
See what is filling the context	/context: a usage grid

THE TIEBREAKER

Clear between unrelated tasks. Old context biases new work, as module 10 showed.

Auto mode

INTERACTIVE SESSIONS

Auto is the starting mode

- A classifier model reviews each action
- Fewer approval prompts for routine work
- Risky actions still stopped or asked
- Default from v2.1.283

CONTROL

Who decides

- -p and SDK runs start in Manual mode
- Admins can disable it in managed settings
- Project settings cannot switch auto on
- Other modes: acceptEdits, plan, dontAsk, bypass

ON THE EXAM

Auto mode is a classifier, not a guarantee. Hooks and deny rules still enforce what must never happen.

Headless in CI

```
1 # .github/workflows/review.yml (step)
2 claude -p "Review this diff against CLAUDE.md conventions. List blocking issues." \
3   --bare \
4   --output-format json \
5   --json-schema review_schema.json \
6   --allowedTools "Read,Grep,Glob" \
7   --permission-mode dontAsk \
8   > review.json
9 python ci/post_review.py review.json       # code decides: comment, or fail the job
```

3-5 Bare mode for scripts; JSON validated against a schema.

6-7, 9 Read-only tools, no prompts; code acts on the result.

CURRENCY CHECK

--bare is "the recommended mode for scripted and SDK calls". Streaming JSON is there for live progress.

Copperlet: hooks on everything

BEFORE

One slow hook, every call

- PreToolUse matched every tool, including reads
- A 9-second security scan each time
- Sessions painfully slow
- Engineers disabled hooks locally

AFTER

The right hook, the right place

- PreToolUse on Edit|Write: protect secrets, fast
- PreToolUse on Bash: block dangerous commands
- PostToolUse on edits: formatter
- Full security scan moved to CI

SETTLED

Quirk 14 settled: match hooks narrowly, keep them fast, and put slow checks where waiting is normal.

> A team's CLAUDE.md contains a long section on database migration conventions. It loads in every session, though only the migrations folder needs it, and it crowds the context.
>
> The rest of the file is short.

What should the team do?

A Delete the migration conventions to save context

B Move them into a rule scoped with paths: to the migrations folder

C Move them into each developer's local CLAUDE.md

D Put them in auto memory instead

◆ PAUSE AND DECIDE

What should the team do?

A Delete the migration conventions to save context

Loses guidance the folder needs.

B Move them into a rule scoped with paths: to the migrations folder

Loads only where it applies.

C Move them into each developer's local CLAUDE.md

Unshared, and still always loaded.

D Put them in auto memory instead

Memory is not reviewed convention.

THE SIGNAL

Guidance for some files only: a path-scoped rule.

> A team has a production deploy procedure as a Skill. Claude sometimes starts it on its own when a conversation mentions deploying.
>
> Deploys must only start when a person explicitly asks.

What should the team change?

A. Shorten the Skill's description so it matches less often

B. Move the procedure into CLAUDE.md

C. Ask Claude in the Skill body to wait for permission

D. Set disable-model-invocation on the Skill

◆ PAUSE AND DECIDE

What should the team change?

A Shorten the Skill's description so it matches less often

Less often is not never.

B Move the procedure into CLAUDE.md

Loads every session; still no gate.

C Ask Claude in the Skill body to wait for permission

The body loads after triggering.

D Set disable-model-invocation on the Skill

User-only, by configuration.

THE SIGNAL

"Only when a person asks" is the setting disable-model-invocation.

> A CI job runs Claude Code to label pull requests. It sometimes hangs until the job times out, and the team's parser struggles with free-form text output.
>
> The job needs no files written.

What should the job use?

A claude -p with --bare, schema-checked JSON and read-only tools

B An interactive session started by the CI runner

C bypassPermissions, so nothing ever waits

D A longer job timeout and a smarter text parser

◆ PAUSE AND DECIDE

What should the job use?

A claude -p with --bare, schema-checked JSON and read-only tools

No prompts to hang on; parseable output.

B An interactive session started by the CI runner

Nobody is there to answer.

C bypassPermissions, so nothing ever waits

Allows every tool, including writes.

D A longer job timeout and a smarter text parser

Treats both symptoms, fixes neither.

THE SIGNAL

Hangs mean an approval prompt nobody answers. Free text means no schema.

> After a two-hour session refactoring the billing module, a developer starts on an unrelated documentation task in the same session. Claude keeps referring to billing details and suggests billing-style patterns.
>
> The developer wants a clean start.

What should the developer do?

A Run /compact, keeping the billing summary

B Run /clear before starting the unrelated task

C Add a CLAUDE.md line saying to ignore billing

D Resume the billing session again with --resume

◆ PAUSE AND DECIDE

What should the developer do?

A Run /compact, keeping the billing summary
Keeps the very context that biases it.

B Run /clear before starting the unrelated task
An empty context for new work.

C Add a CLAUDE.md line saying to ignore billing
Affects every future session.

D Resume the billing session again with --resume
More of the same context.

THE SIGNAL

Unrelated new task: clear. Same task, crowded context: compact.

> An engineering team rolls out Claude Code with a PreToolUse hook matching every tool. It runs a slow scan, and sessions crawl. Engineers start disabling hooks. Security still needs secrets protected and dangerous commands blocked.
>
> The team wants both speed and safety.

Which two changes fit best? Choose two.

A. Remove all hooks, and rely on CLAUDE.md for the rules

B. Keep the hook on every tool, but run the scan in the background

C. Match hooks narrowly, on edits and Bash, with fast checks

D. Let each engineer choose which hooks to keep locally

E. Move the slow full scan into CI, where waiting is expected

◆ PAUSE AND DECIDE

Which two changes fit best? Choose two.

A Remove all hooks, and rely on CLAUDE.md for the rules

Loses the enforcement security needs.

B Keep the hook on every tool, but run the scan in the background

A background scan cannot block.

C Match hooks narrowly, on edits and Bash, with fast checks

Enforcement only where it matters.

D Let each engineer choose which hooks to keep locally

Safety becomes optional.

E Move the slow full scan into CI, where waiting is expected

Thorough, without slowing sessions.

THE SIGNAL

Narrow, fast hooks for what must never happen; slow checks in CI. Copperlet's quirk 14.

> A regulated company must ensure that Claude Code never runs in auto mode on any developer's machine. A team lead adds a setting to the shared project settings file to turn auto mode off.
>
> Some developers still see auto mode.

What should the company do?

A Add a CLAUDE.md line saying auto mode is not allowed

B Ask developers to switch modes manually each session

C Use headless mode only, for all developers

D Disable auto mode in managed settings

◆ PAUSE AND DECIDE

What should the company do?

A Add a CLAUDE.md line saying auto mode is not allowed

Context cannot change the mode.

B Ask developers to switch modes manually each session

Relies on memory, every day.

C Use headless mode only, for all developers

Removes interactive use entirely.

D Disable auto mode in managed settings

The documented organisation control.

CURRENCY CHECK

"Never, on any machine" is a managed setting: permissions.disableAutoMode.

Module 27 in five lines

- **Components:** CLAUDE.md, path rules, Skills, subagents, agent memory; settings and hooks enforce.
- **Commands are Skills now;** disable-model-invocation for user-only actions.
- **Sessions:** /init to start, /clear for new work, /compact when crowded, --resume to return.
- **Auto mode** starts interactive sessions; -p and SDK start in Manual; managed settings decide.
- **CI:** -p, --bare, JSON schema, read-only tools; narrow, fast hooks.

COMING UP

Next, Part 8: security - the AI-specific threat model.